Diabetes Prediction from Clinical Measurements

A comparative machine learning study on the Pima Indians Diabetes dataset

Course	CBIO313 - Data Mining & Machine Learning
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Date	17 August 2026
Repository	github.com/ssameh12/diabetes-prediction-cbio313
Video demo	https://drive.google.com/file/d/1uIBeFOJZGFTf5Zkxpa6KXynUFkqJWbOX/view?usp=drive_link

Abstract

Type 2 diabetes is frequently asymptomatic until complications appear, which makes cheap triage valuable. This study builds a screening classifier from eight routine clinical and demographic measurements on 768 patients. The central preprocessing finding is that missing measurements in this dataset were silently recorded as zeros - affecting 48.7% of insulin and 29.6% of skin-fold values - and correcting this is more consequential than the choice of algorithm. Five algorithms were compared under identical preprocessing; a tuned SVM (RBF) was selected on cross-validated ROC-AUC (0.849) computed on training data alone, reaching a test ROC-AUC of 0.810. Moving the decision threshold from 0.50 to 0.28, chosen from out-of-fold training predictions, raised recall on diabetic patients from 50.0% to 81.5% at a cost of only 2.9% precision - reducing missed diabetics from 27 to 10 out of 54.

1. Introduction and problem definition

Type 2 diabetes develops silently. By the time symptoms prompt a clinic visit, damage to the kidneys, retina and peripheral nerves is often already underway. Screening programmes address this, but a full diagnostic workup cannot be offered to an entire population. A model that flags who most needs that workup, using measurements already collected during a routine visit, is therefore practically useful.

Objectives.

- Correct a dataset in which missing values were recorded as zeros.
- Characterise how each measurement relates to diabetes status.
- Train and compare five classification algorithms covered in the course.
- Evaluate with metrics suited to an imbalanced medical screening task, not accuracy alone.
- Tune the strongest model and select an operating threshold appropriate to screening.
- Explain the model's predictions so that a clinician could interrogate them.

The metric that matters. The two error types are not equally costly. A false positive sends a healthy patient for a confirmatory test: inconvenient and mildly expensive. A false negative sends an undiagnosed diabetic home reassured. Recall on the positive class is therefore the primary metric, with precision monitored so the model does not degenerate into flagging everyone.

2. Dataset description

Source. National Institute of Diabetes and Digestive and Kidney Diseases, distributed through the UCI Machine Learning Repository as the Pima Indians Diabetes Database. **Population.** 768 women of Pima Indian heritage, aged 21 or older, living near Phoenix, Arizona. 8 predictive features and one binary target. Class distribution: 500 without diabetes and 268 with, a positive rate of 34.9%.

Feature	Description	Correlation with outcome
Pregnancies	Number of times pregnant	+0.222
Glucose	Plasma glucose, 2 h OGTT (mg/dL)	+0.495
BloodPressure	Diastolic blood pressure (mm Hg)	+0.171
SkinThickness	Triceps skin-fold thickness (mm)	+0.259
Insulin	2-hour serum insulin (mu U/mL)	+0.303
BMI	Body mass index (kg/m^2)	+0.314
DiabetesPedigreeFunction	Family-history diabetes score	+0.174
Age	Age (years)	+0.238

3. Data preprocessing

3.1 Missing values disguised as zeros

Five columns report a minimum of zero: Glucose, BloodPressure, SkinThickness, Insulin and BMI. None is biologically possible in a living patient. These are unrecorded measurements that were never encoded as missing. Left uncorrected, a model would read *insulin = 0* as a genuine and extremely low reading, which is the opposite of the truth.

Feature	Zeros	% of rows
Insulin	374	48.7%
SkinThickness	227	29.56%
BloodPressure	35	4.56%
BMI	11	1.43%
Glucose	5	0.65%

Dropping incomplete rows was rejected: it would discard roughly half the dataset, and patients missing an insulin measurement are unlikely to be a random sample. The zeros were converted to NaN and filled with the median, which is robust to the pronounced right skew of these variables.

3.2 Preventing data leakage

Imputation and scaling are placed **inside a scikit-learn Pipeline** rather than applied to the full dataset beforehand. Computing a median over all rows and only then splitting would let information about test patients influence the training data, inflating every score that follows. Within a pipeline, both are refitted from the training portion of each cross-validation fold. The split is stratified 80/20: 614 training and 154 test patients, preserving the class ratio in both.

4. Exploratory data analysis

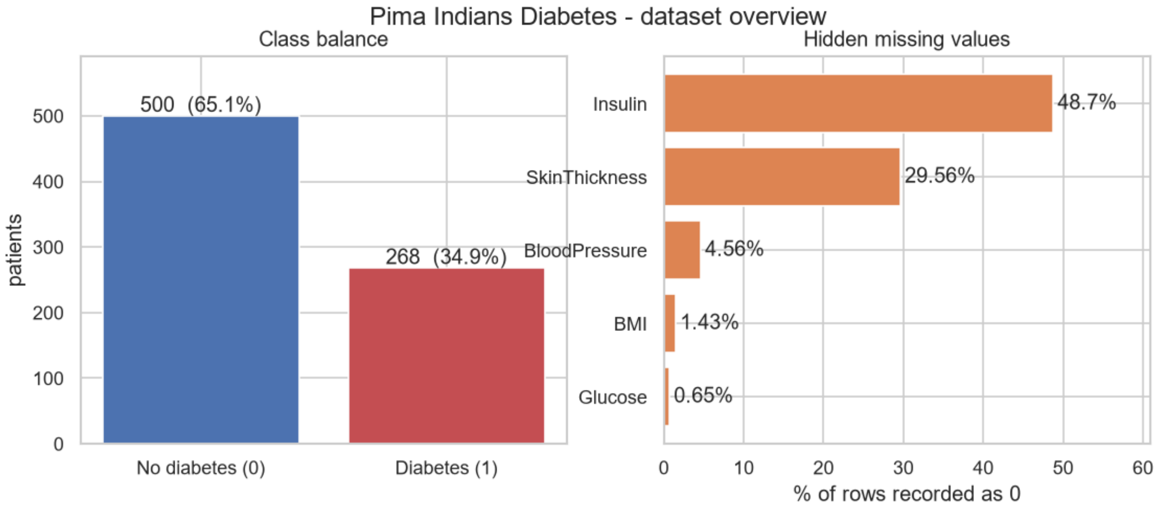

Figure 1. Class balance and the extent of the hidden missing values.

The classes are imbalanced at roughly 65% / 35%. A model predicting 'no diabetes' for every patient would score 65.1% accuracy while catching no diabetics at all. That is the benchmark any headline accuracy must be read against.

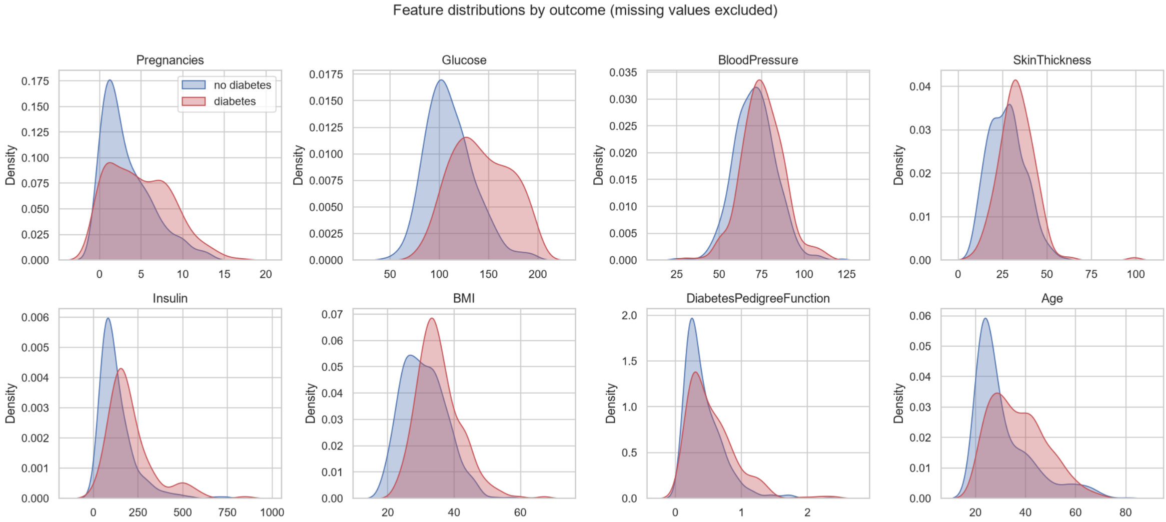

Figure 2. Feature distributions split by outcome, missing values excluded.

Glucose separates the groups most clearly, consistent with its correlation of +0.49 and with plasma glucose forming part of the diagnostic definition of diabetes. BMI (+0.31) and Age (+0.24) shift more modestly. BloodPressure and SkinThickness overlap almost entirely, anticipating their negligible contribution in the importance analysis.

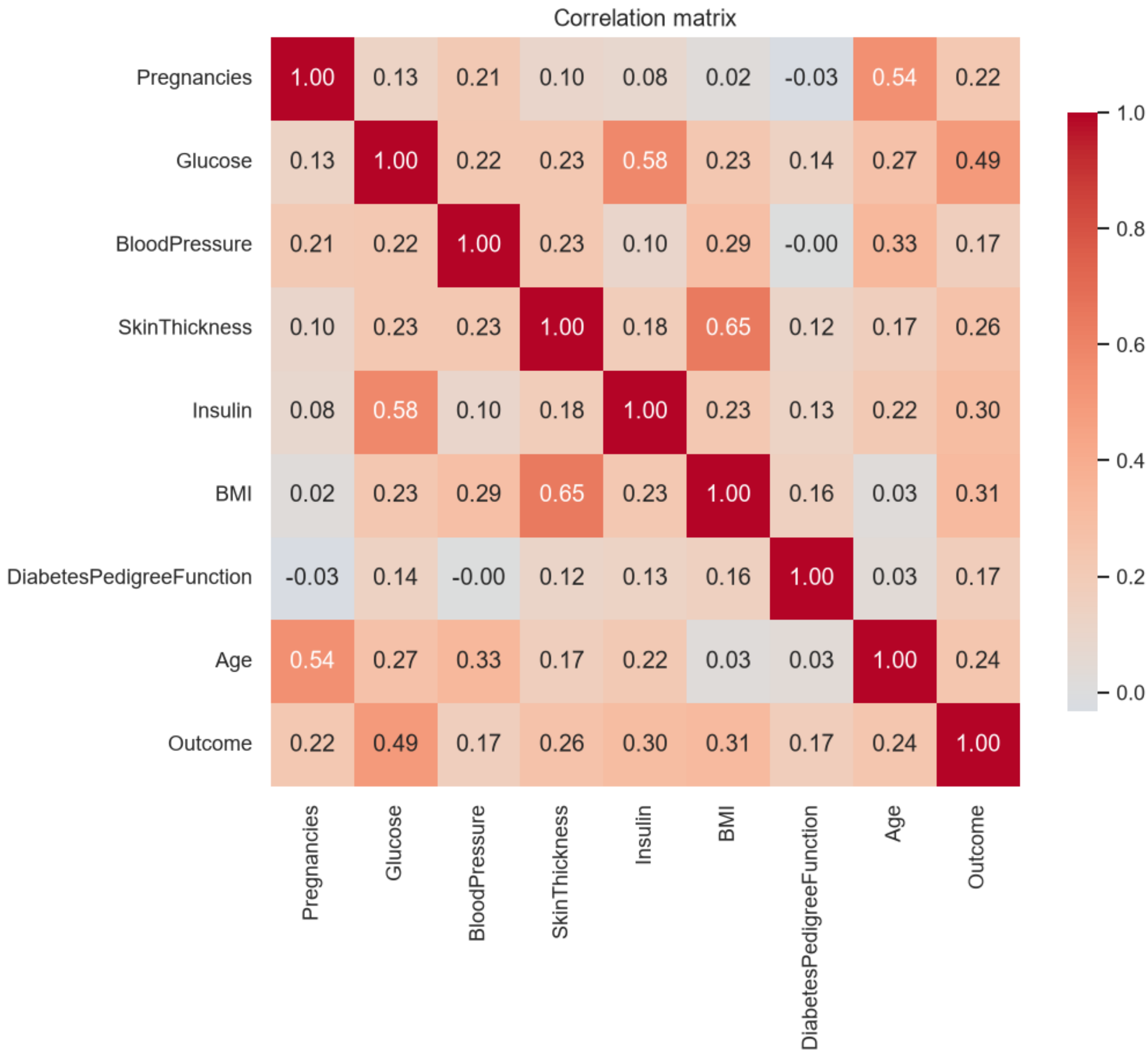

Figure 3. Correlation matrix of features and outcome.

Age correlates with Pregnancies, and SkinThickness with BMI. This collinearity means linear coefficients should not be read as independent effects, which is one reason the SHAP analysis in section 7 carries more weight than raw coefficients.

5. Machine learning implementation

Five algorithms from the course syllabus, each wrapped in the identical impute-scale-classify pipeline so that the comparison isolates the algorithm: Logistic Regression (linear baseline), K-Nearest Neighbours (non-parametric, local), Decision Tree (interpretable rules), Random Forest (bagged ensemble) and a Support Vector Machine with an RBF kernel (maximum margin, non-linear).

5.1 Cross-validated comparison

A single split of 768 rows is noisy, so models are ranked by stratified 5-fold cross-validation on the training set, leaving the test set untouched for the final estimate.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	**0.788**	**0.764**	**0.575**	**0.654**	**0.843**
SVM (RBF)	0.780	0.742	0.584	0.648	0.833
K-Nearest Neighbours	0.752	0.690	0.556	0.611	0.822
Random Forest	0.769	0.700	0.603	0.646	0.821
Decision Tree	0.673	0.531	0.580	0.552	0.651

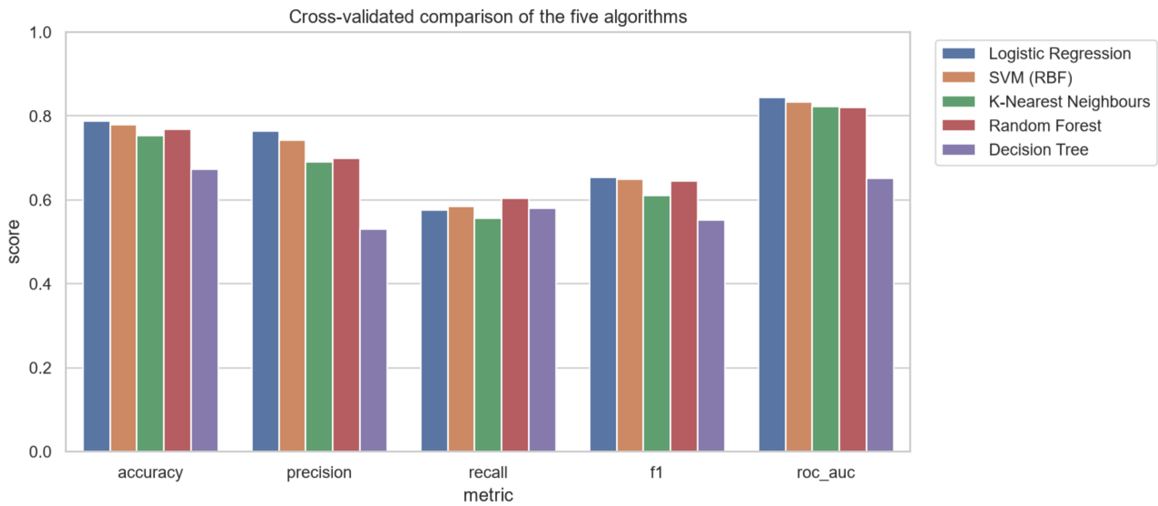

Figure 4. Cross-validated metrics for the five algorithms.

Recall sits far below accuracy for every model - at the default threshold these classifiers detect only about 55 to 60 percent of diabetics. Section 6.2 addresses this directly.

6. Evaluation and comparison

6.1 Held-out test set

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.740	0.659	0.537	0.592	0.816
Logistic Regression	0.708	0.600	0.500	0.545	0.813
SVM (RBF)	0.740	0.652	0.556	0.600	0.796
K-Nearest Neighbours	0.734	0.633	0.574	0.602	0.788
Decision Tree	0.682	0.553	0.481	0.515	0.636

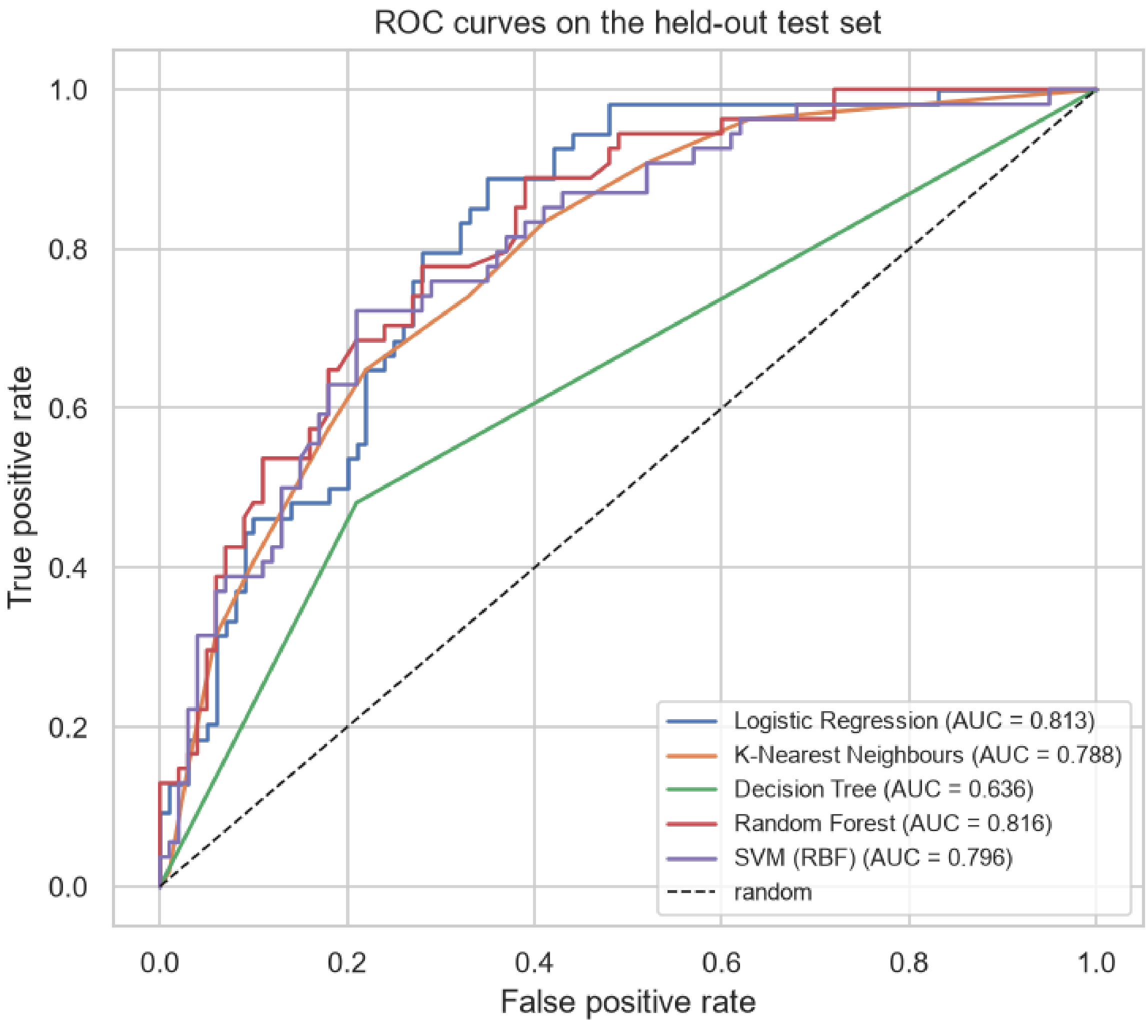

Figure 5. ROC curves on the held-out test set.

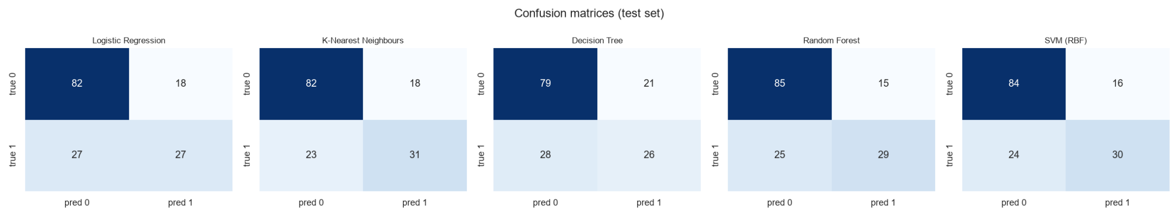

Figure 6. Confusion matrices at the default 0.50 threshold. The lower-left cell of each is the clinically costly error: diabetics the model sent home.

6.2 Hyperparameter tuning and threshold selection

Grids were scored on ROC-AUC rather than recall. Optimising a grid directly on recall rewards models that predict the positive class more often, which collapses precision; in the limit, predicting everyone positive scores perfect recall. ROC-AUC ranks models independently of the threshold, and the threshold is then chosen as a separate decision.

Model	Best parameters	CV ROC-AUC	Test ROC-AUC
Logistic Regression	C=0.1, class_weight=balanced	0.8450	0.8104
Random Forest	class_weight=None, max_depth=5, min_samples_leaf=3, n_estimators=400	0.8388	0.8119
SVM (RBF)	C=1, class_weight=balanced, gamma=0.01	0.8493	0.8096

SVM (RBF) was selected on cross-validated ROC-AUC (0.8493) with parameters C=1, class_weight=balanced, gamma=0.01. Selecting by test score would have leaked the test set into model selection.

The default 0.50 cut-off maximises accuracy, the wrong target here. A threshold of **0.285** was chosen as the highest value still reaching 80% recall on out-of-fold training predictions, so the test set played no part in the choice.

Operating point	Accuracy	Precision	Recall	F1	Missed diabetics
Default threshold 0.50	0.708	0.600	0.500	0.545	27 of 54
Chosen threshold 0.285	**0.721**	**0.571**	**0.815**	**0.672**	**10 of 54**

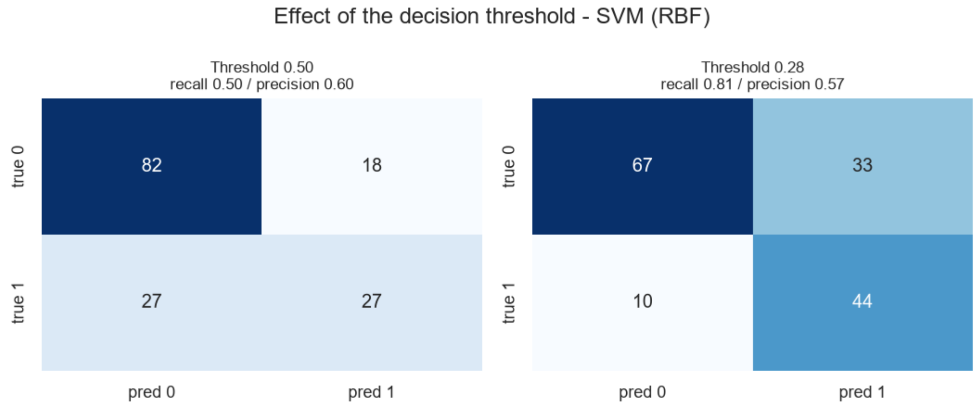

Figure 7. Confusion matrices at the two operating points.

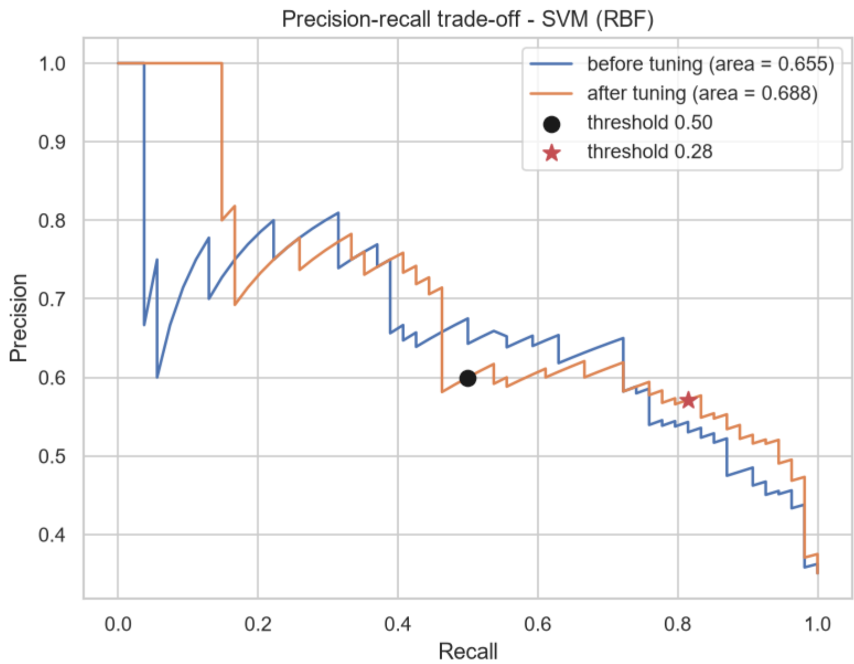

Figure 8. Precision-recall trade-off, with both operating points marked.

7. Explainability

Two complementary views were produced: permutation importance, which shuffles one feature and measures the resulting drop in ROC-AUC on held-out data, and SHAP, which attributes each individual prediction to its features.

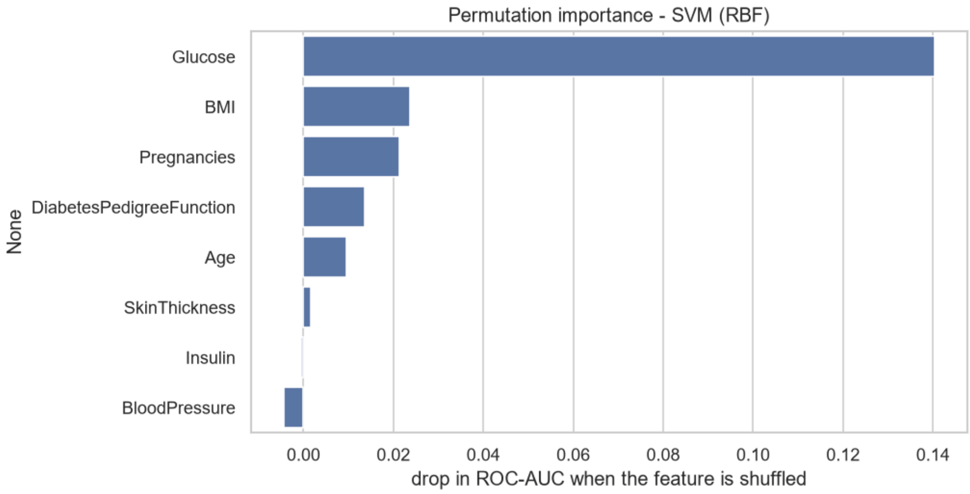

Figure 9. Permutation importance on the test set.

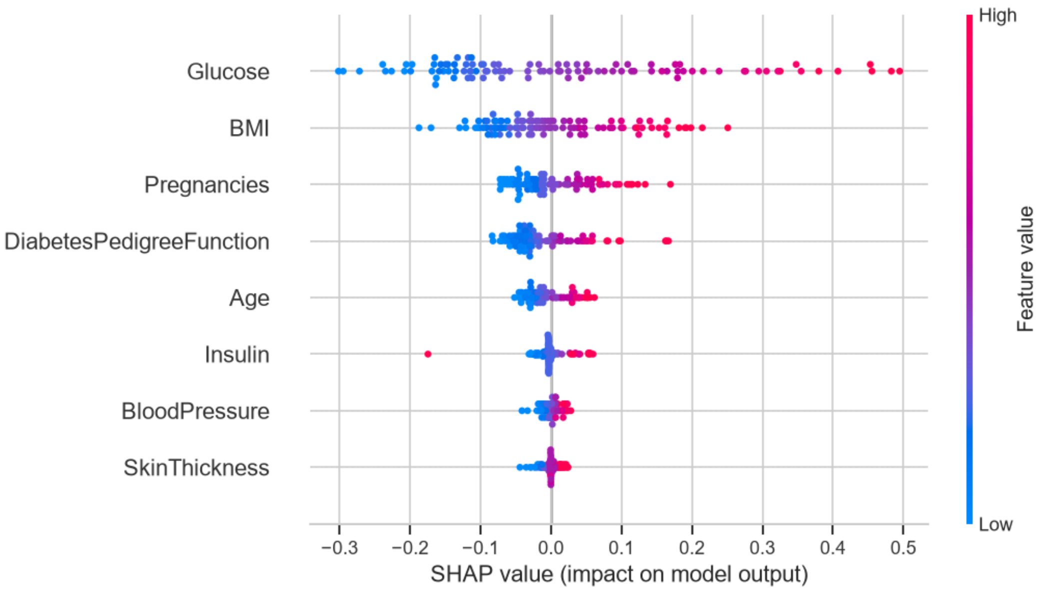

Figure 10. SHAP summary. Each dot is one patient; horizontal position is the push towards a diabetes prediction, colour is whether that feature's value was high (red) or low (blue).

Both methods agree on the ordering, led by Glucose, then BMI and Pregnancies. Critically, the directions are clinically correct: high glucose and high BMI push predictions towards diabetes. A model with acceptable accuracy but an inverted relationship would be dangerous, and this plot is what would expose it.

8. Discussion and conclusion

A tuned SVM (RBF) reached a test ROC-AUC of 0.810, in line with published results on this dataset. Logistic Regression finished within a fraction of it, which is worth stating plainly: on 614 training rows with eight features, a well-regularised linear model is competitive with anything more elaborate. The single decision tree was the clear laggard.

The decisive result is the threshold. Moving it from 0.50 to 0.28 converted a model that missed 27 of 54 diabetics into one that misses 10, lifting recall from 50.0% to 81.5%. Accuracy barely moved (0.708 to 0.721) while recall gained 31 points, which is the clearest possible demonstration that accuracy was not measuring the quantity of interest. The gains in this project came from preprocessing and from matching the operating point to the clinical cost of each error - not from the choice of algorithm, where four of five models landed within a few points of one another.

Limitations

- **Narrow population.** 768 patients, all women of Pima heritage aged 21 and over near Phoenix. Nothing here licenses use on men, other ethnic groups, or younger patients; the unusually high local prevalence means even the base rate does not transfer.
- **Heavy imputation.** 48.7% of insulin and 29.6% of skin-fold values were filled with the median. Those two features are correspondingly the least trustworthy - and rank near the bottom of the importance plots.
- **Missingness is likely not random.** A patient with no recorded insulin value probably differs systematically from one who has it; median imputation assumes otherwise.
- **No external validation.** All results derive from one 154-patient test split of a single dataset, so the interval around the reported recall is wide.
- **Information ceiling.** Eight coarse measurements carry limited signal. Materially better results need better features - HbA1c, fasting glucose, waist circumference - not more elaborate models.

Conclusion

As a triage aid flagging roughly 81% of diabetics for confirmatory testing, the model is plausible and its behaviour is explainable. As a diagnostic device it is not, for the reasons above. Future work: external validation on a broader cohort, multiple imputation with an explicit missing-indicator feature, probability calibration so outputs can be read as real risks, and cost-sensitive learning driven by the actual cost of each error type.

References

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